Thesis Improvement Update

Habituation-Aware Multi-Robot Intervention

1 Background and scope

Plans A–E were implemented as analytical improvements to the STL predictive value function: arrival-rate forecast correction (B), integrated exposure scoring (B_ie), forward trajectory correction (C), full greedy continuation rollout (D), and reactive-task service protection (E). Each plan was evaluated against the B4 baseline (variant A) across 30 seeds at three load regimes. This document reports the experimental outcomes, the structural diagnosis they support, and a revised improvement proposal targeting the actual performance bottlenecks.

2 Experimental results: Plans A–E

All values are mean value-weighted exposure relative to variant A (the B4 baseline), with habituation enabled ($\kappa = 0.50$, $n = 30$ seeds, $T = 1800\,\mathrm{s}$). Differences within $\pm 0.5\%$ are within measurement noise at this sample size.

Variant	Added over B4 baseline	μ=2×10^{-5}	μ=1×10^{-4}	μ=4×10^{-4}	Verdict
A (B4 baseline)	Reference	0.0%	0.0%	0.0%	Reference
B	Arrival-rate forecast correction	+0.1%	+0.0%	−0.0%	Null
B_ie	Integrated exposure scoring	+0.3%	+0.1%	−0.0%	Slight regression
C	B + forward trajectory correction	+0.1%	+0.0%	−0.0%	Null
C_ie	C + integrated exposure	+0.4%	+0.1%	−0.0%	Slight regression
D	Greedy continuation rollout	+0.1%	−0.1%	+0.0%	Null
E	D + service protection gate	+0.1%	−0.1%	+0.0%	E = D (correct)

3 Diagnosis

The value function is not the bottleneck. Plans A–E collectively prove by elimination that improving how accurately the system estimates the marginal value of a predictive visit does not change outcomes. The performance ceiling is structural.

At spare load ($\mu = 2\times10^{-5}$)

Robot capacity exceeds prioritized demand. Nearly every high-value cell is served regardless of which value function ranks the candidates. More accurate estimation is solving the wrong problem.

At heavy and overload ($\mu = 1\times10^{-4}$, 4×10^{-4})

Robots are increasingly consumed by reactive deterrence (responding to actual bird detections). At $\mu = 4\times10^{-4}$, predictive tasks are rarely dispatched at all. A more accurate value function for predictive candidates is irrelevant when there are no free robots to execute them.

The overload reversal (+5.5% vs. B1 at $\mu = 4\times10^{-4}$)

The STL reservation mechanism holds robot-time slots for predictive tasks even when the system is capacity-constrained on reactive demand. This backfires at overload: reserved slots become unavailable for reactive tasks, increasing cumulative exposure above the greedy B1 reference. The value function plays no role in this failure mode.

Why integrated exposure variants slightly hurt

Scoring cells by total accumulated exposure over a window over-weights cells that have already seen high cumulative exposure, which may already be recovering after a recent visit. B4's instantaneous exposure score is a better proxy for marginal deterrence value.

Plan E = Plan D (correct behavior, not a bug)

Service protection is structurally sound but dormant in the tested regime. The fleet pending-events list only includes predictive deterrence tasks (mode $\neq$ None). At $\mu = 4\times10^{-4}$, robots perform reactive-only deterrence (mode = None, excluded from the list), so all robots appear free to the protection filter and it never triggers. The conflict the filter guards against does not arise.

4 Positive finding: preemption

The preemption variant (advisor item 1) is the only tested intervention that measurably reduces exposure. When a reactive task arrives while a robot is mid-way through a predictive visit, preemption aborts the predictive task and responds immediately. The benefit grows with load, confirming the mechanism.

Load	B1 ref.	B4 base	B4+preempt	vs. B4	vs. B1
μ=2×10^{-5} (spare)	40,441	36,006	35,896	−0.3%	−11.2%
μ=1×10^{-4} (heavy)	195,746	185,159	183,664	−0.8%	−6.2%
μ=4×10^{-4} (overload)	775,170	817,880	808,679	−1.1%	+4.3%

Preemption closes part of the overload reversal gap (from +5.5% to +4.3% vs. B1) but does not eliminate it. The remaining gap motivates items 3B and 3C below.

5 Revised proposal: three counter-interventions

The null results redirect the thesis toward structural interventions. Each item below targets a bottleneck identified by the diagnostic above.

3A Fleet size and reactive pressure characterization

Adding robots is the most direct performance lever but is not freely available in practice. What is needed first is a characterization of the deployment space: for a given vineyard configuration and bird arrival rate μ, what is the minimum fleet size N^* such that the STL reservation overhead is justified—i.e., B4 does not reverse vs. B1?

Design. Sweep $n_{\text{robots}} \in \{2, 3, 4, 5, 6\}$ at three loads ($\mu = 2\times10^{-5}$, 1×10^{-4}, 4×10^{-4}), 10 seeds each, with B1 and B4 as comparison systems. Complement with a queuing-theoretic lower bound on N^*. Primary output: a (μ, N) phase diagram showing the boundary where B4 transitions from better to worse than B1. Total: ≈ 900 simulations.

Thesis contribution. Quantitative deployment guidance for vineyard operators: given a known bird pressure and vineyard size, what fleet is required for the STL system to outperform greedy baseline?

3B Heterogeneous fleet with drone integration

The overload reversal arises because ground robots must serve both reactive and predictive demand from a shared pool. A heterogeneous fleet separates these two functions: fast drones handle reactive detections exclusively (shorter travel time, $\approx$12 s vs. 30 s for ground), while ground robots operate in a protected predictive-only regime under the existing STL scheduler. This should eliminate the reservation-vs.-reactive conflict.

Design. Test six configurations at three loads, holding total agent count at four and varying the ground/drone split. Primary comparison: four ground (baseline) vs. three ground + one drone vs. two ground + two drones, with and without preemption. Total: $\approx$360 simulations.

Implementation. Add a drone robot type to `DeterrentSystem.py` with separate travel and service hold parameters. Drones receive reactive tasks only; the STL planner counts ground robots only. No changes to the scheduler logic are required.

Expected outcome. At $\mu = 4\times10^{-4}$, the 3-ground +1-drone configuration is expected to eliminate or substantially reduce the +5.5% reversal. At spare load, the marginal reduction in ground-robot predictive capacity may slightly reduce the B4 vs. B1 gain; the net tradeoff determines whether heterogeneous deployment is Pareto-dominant across loads.

3C Load-aware dispatch gate

A simple feedback controller detects when reactive task backlog per robot exceeds a threshold and temporarily suspends STL predictive scheduling, releasing reserved robot-slots back to reactive demand. This directly addresses the mechanism of the overload reversal without changing the fleet configuration:

$$\text{reactive pressure} \;=\; \frac{|\text{pending reactive tasks}|}{n_{\text{robots}}} \;\geq\; \theta \;\Rightarrow\; \text{suspend STL this step.}$$

When pressure drops below θ, predictive scheduling resumes normally.

Design. Threshold sweep at $\mu = 4\times10^{-4}$ (5 seeds) to select θ, then full comparison across all loads with four systems: B4 baseline, B4 + preempt, B4 + load-aware, and B4 + preempt + load-aware. Run alongside item 3B for direct side-by-side comparison. Total: $\approx$300 simulations.

Implementation. One new parameter in `DeterrentSystem.py`; one conditional in the per-step dispatch loop. Compatible with all existing configurations.

6 Summary of empirical reference values

Quantity	Value
B4 vs. B1 at spare load (hab. on, $n = 30$)	-11.0%
B4 vs. B1 at heavy load (hab. on, $n = 30$)	-5.4%
B4 vs. B1 at overload (hab. on, $n = 30$)	$+5.5\%$ (reversal)
B4+preempt vs. B4 at spare load ($n = 30$)	-0.3%
B4+preempt vs. B4 at heavy load ($n = 30$)	-0.8%
B4+preempt vs. B4 at overload ($n = 30$)	-1.1%
B4+preempt vs. B1 at overload ($n = 30$)	$+4.3\%$ (reversal persists)
Maximum Plans B–E change vs. A (any load)	0.4% (within noise)
Ground robot avg. travel time	30 s
Ground robot service hold time	20 s
Proposed drone avg. travel time	12 s
Fleet size (current)	4

7 Request for feedback

Before beginning implementation of items 3A–3C, I wanted to confirm that this direction is consistent with the scope of the thesis and that the proposed experiments are appropriately sized. Three specific questions:

(1) Is the fleet size characterization (3A) appropriate as a thesis contribution, or better framed as future work with a brief analytical argument?

(2) Is drone integration (3B) realistic within the thesis scope, or should it be framed as a simulation-only feasibility study with a note on hardware assumptions?

(3) Is the load-aware gate (3C) sufficient on its own to address the overload reversal, or is the full three-item agenda needed to make a complete thesis chapter?

Happy to discuss at your convenience.